Binary Search Tree (BST) Traversals – Inorder, Preorder, Post Order

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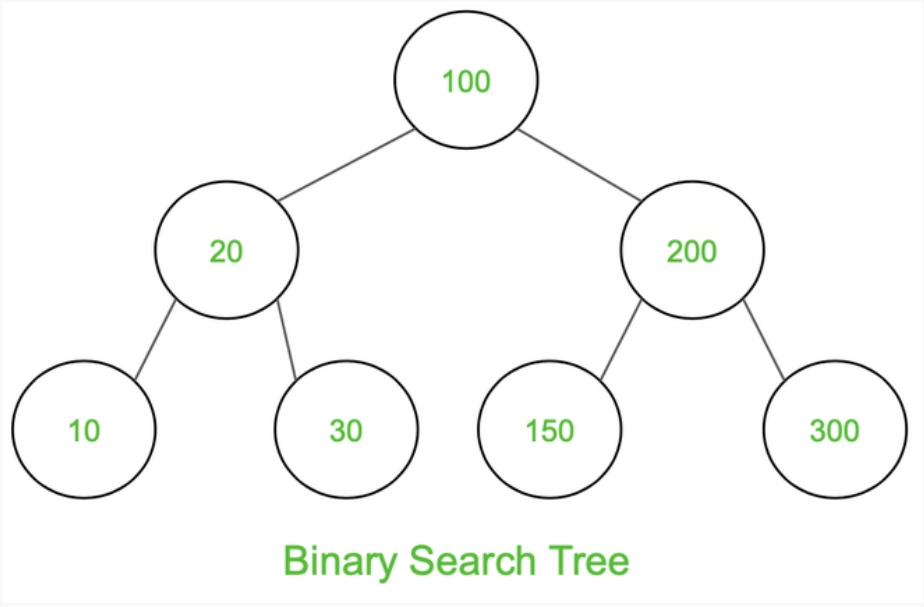
Given a Binary Search Tree, The task is to print the elements in inorder, preorder, and postorder traversal of the Binary Search Tree.

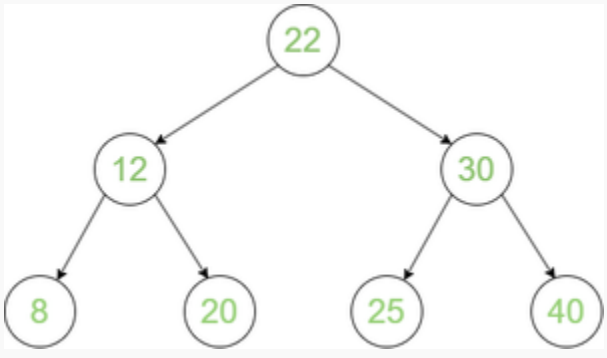
Input:



A Binary Search Tree

Output:
Inorder Traversal: 10 20 30 100 150 200 300
Preorder Traversal: 100 20 10 30 200 150 300
Postorder Traversal: 10 30 20 150 300 200 100

Input:



Binary Search Tree

Output:
Inorder Traversal: 8 12 20 22 25 30 40
Preorder Traversal: 22 12 8 20 30 25 40
Postorder Traversal: 8 20 12 25 40 30 22

Inorder Traversal:

Below is the idea to solve the problem:

> *At first traverse **left subtree** then visit the **root** and then traverse the **right subtree**.*

Follow the below steps to implement the idea:

- Traverse left subtree
- Visit the root and print the data.
- Traverse the right subtree

The inorder traversal of the BST gives the values of the nodes in sorted order. To get the decreasing order visit the right, root, and left subtree.

Below is the implementation of the inorder traversal.

C++ Java Python C# JavaScript

```cpp
// C++ code to implement the approach

#include <bits/stdc++.h>
using namespace std;

// Class describing a node of tree
class Node {
public:
    int data;
    Node* left;
    Node* right;
    Node(int v)
    {
        this->data = v;
        this->left = this->right = NULL;
    }
};

// Inorder Traversal
void printInorder(Node* node)
{
    if (node == NULL)
```

```
        return;

    // Traverse left subtree
    printInorder(node->left);

    // Visit node
    cout << node->data << " ";

    // Traverse right subtree
    printInorder(node->right);
}

// Driver code
int main()
{
    // Build the tree
    Node* root = new Node(100);
    root->left = new Node(20);
    root->right = new Node(200);
    root->left->left = new Node(10);
    root->left->right = new Node(30);
    root->right->left = new Node(150);
    root->right->right = new Node(300);

    // Function call
    cout << "Inorder Traversal: ";
    printInorder(root);
    return 0;
}
```

Output

```
Inorder Traversal: 10 20 30 100 150 200 300
```

Time complexity: O(N), Where N is the number of nodes.

Auxiliary Space: O(h), Where h is the height of tree

Preorder Traversal:

Below is the idea to solve the problem:

*At first visit the **root** then traverse **left subtree** and then traverse the **right subtree**.*

Follow the below steps to implement the idea:

- Visit the root and print the data.
- Traverse left subtree
- Traverse the right subtree

Below is the implementation of the preorder traversal.

C++ Java Python C# JavaScript

```cpp
// C++ code to implement the approach

#include <bits/stdc++.h>
using namespace std;

// Class describing a node of tree
class Node {
public:
    int data;
    Node* left;
    Node* right;
    Node(int v)
    {
        this->data = v;
        this->left = this->right = NULL;
    }
};

// Preorder Traversal
void printPreOrder(Node* node)
{
    if (node == NULL)
        return;

    // Visit Node
    cout << node->data << " ";

    // Traverse left subtree
```

```
    printPreOrder(node->left);

    // Traverse right subtree
    printPreOrder(node->right);
}

// Driver code
int main()
{
    // Build the tree
    Node* root = new Node(100);
    root->left = new Node(20);
    root->right = new Node(200);
    root->left->left = new Node(10);
    root->left->right = new Node(30);
    root->right->left = new Node(150);
    root->right->right = new Node(300);

    // Function call
    cout << "Preorder Traversal: ";
    printPreOrder(root);
    return 0;
}
```

Output

```
Preorder Traversal: 100 20 10 30 200 150 300
```

Time complexity: O(N), Where N is the number of nodes.

Auxiliary Space: O(H), Where H is the height of the tree

Postorder Traversal:

Below is the idea to solve the problem:

> *At first traverse **left subtree** then traverse the **right subtree** and then visit the **root**.*

Follow the below steps to implement the idea:

- Traverse left subtree

- Traverse the right subtree
- Visit the root and print the data.

Below is the implementation of the postorder traversal:

C++ Java Python C# JavaScript

```cpp
// C++ code to implement the approach

#include <bits/stdc++.h>
using namespace std;

// Class to define structure of a node
class Node {
public:
    int data;
    Node* left;
    Node* right;
    Node(int v)
    {
        this->data = v;
        this->left = this->right = NULL;
    }
};

// PostOrder Traversal
void printPostOrder(Node* node)
{
    if (node == NULL)
        return;

    // Traverse left subtree
    printPostOrder(node->left);

    // Traverse right subtree
    printPostOrder(node->right);

    // Visit node
    cout << node->data << " ";
}
```

```
// Driver code
int main()
{
    Node* root = new Node(100);
    root->left = new Node(20);
    root->right = new Node(200);
    root->left->left = new Node(10);
    root->left->right = new Node(30);
    root->right->left = new Node(150);
    root->right->right = new Node(300);

    // Function call
    cout << "PostOrder Traversal: ";
    printPostOrder(root);
    cout << "\n";

    return 0;
}
```

Output

```
PostOrder Traversal: 10 30 20 150 300 200 100
```

Time complexity: O(N), Where N is the number of nodes.

Auxiliary Space: O(H), Where H is the height of the tree

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